

The Effects of Commercial Preparations of Red Raspberry Leaf on the Contractility of the Rat's Uterus In Vitro

Jing Zheng, Marc J. Pistilli, Alison C. Holloway, PhD, and
Denis J. Crankshaw, PhD

We investigated the direct effects of various commercially available preparations of red raspberry leaf (RRL) on the in vitro contractility of uteri collected from diethylstilbestrol (DES)-treated nonpregnant (NP) and late pregnant rats. In DES-treated NP rats, RRL tea and capsule caused weak contractions. Neither preparation affected the ability of oxytocin to initiate contractions; however, both partially inhibited preexisting oxytocin-driven contractions at the highest concentration tested. Red raspberry leaf ethanol extract had little effect on contractility. Pretreatment with tea did not alter the ability of oxytocin to initiate contractions. In pregnant animals red raspberry leaf tea had variable effects on preexisting oxytocin-induced contractions, sometimes augmenting oxytocin's effect and sometimes causing augmentation followed by inhibition. We conclude that the biological activity of RRL varies depending on the herbal preparation used and pregnancy status. These results do not support the hypothesis that RRL augments labor by a direct effect on uterine contractility.

KEY WORDS: Uterine contractility, rat, pregnancy, labor, herbal, red raspberry leaf.

INTRODUCTION

There has been a rapid increase in the use of herbal medicines in North America, by all segments of the population including pregnant women.¹⁻⁴ Indeed, it has been estimated that as many as 60% of women use natural health products during pregnancy.⁵⁻⁹ Many of the herbal products consumed during pregnancy are the same as those consumed by the general populace^{10,11}; however, there is a subset of products specifically targeting pregnant women. Among these products, teas, capsules, and extracts made from or containing red raspberry leaf (RRL; *Rubus idaeus*) are heavily promoted for use during

all stages of pregnancy and lactation, including by many nurse-midwives for use during labor.¹²

It has been claimed that administration of RRL capsule in late pregnancy can shorten the second stage of labor and decrease the incidence of forceps delivery,¹³ though the first claim is not supported by the data presented. The lay literature suggests that the beneficial effects of raspberry leaf during labor are due to its ability to strengthen and tone the uterus. It has also been suggested that this effect might be produced by a direct action of RRL on the uterus to coordinate contractions.¹⁴ Although several attempts to test this hypothesis have been made over the years, the results have been inconclusive. Indeed, both excitatory and inhibitory effects of RRL on uterine contractions have been reported,¹⁴⁻¹⁶ but in no case have the effects been adequately quantified. The discrepancies between studies may be attributable to differences in experimental protocols or the different preparations of RRL used. The goal of this study was to test the effects of 3 commercially available forms of RRL (tea, capsule, and ethanol extract) on the in vitro contractility of 2 standardized preparations of the rat's uterus: diethylstilbestrol (DES)-treated nonpregnant (NP)¹⁷ and pregnant animals at gestational day

From the Reproductive Biology Division, Department of Obstetrics and Gynecology, McMaster University, Hamilton, Ontario, Canada L8N 3Z5 (JZ, MJP, ACH, DJC); and Honours Biology and Pharmacology Program, McMaster University, Hamilton, Ontario, Canada L8N 3Z5 (JZ, MJP, DJC).

Address correspondence to: Alison C. Holloway, PhD, Department of Obstetrics and Gynecology, McMaster University, Rm 3N52 HSC, 1200 Main Street W, Hamilton, Ontario, Canada L8N 3Z5. Email: holloway@mcmaster.ca.

Reproductive Sciences Vol. 17 No. 5 May 2010 494-501
DOI: 10.1177/1933719109359703
© 2010 The Author(s)

20 (GD 20; term 21–23 days). We also investigated whether RRL exposure could affect the ability of oxytocin to induce contractions and whether preexisting oxytocin-induced contractions could be modified by subsequent RRL treatment.

MATERIALS AND METHODS

Animal Use

All animal experiments were approved by the Animal Research Ethics Board at McMaster University, in accordance with the guidelines of the Canadian Council for Animal Care. Nulliparous 200 to 250 g female Wistar rats (Harlan, Indianapolis, Indiana) were maintained under controlled lighting (12:12 L:D) and temperature (22°C) with ad libitum access to food and water.

Diethylstilbestrol-Treated NP Uterine Preparations

Females were injected subcutaneously with 0.8 mg/kg DES (Sigma-Aldrich, St Louis, Missouri) in corn oil on each of 2 days prior to sacrifice to produce an estrogen-dominated state. On the third day, animals were sacrificed by CO₂ asphyxiation, and uteri were immediately extracted, placed in physiological salt solution (PSS) containing (in mmol/L): KCl (4.6); MgSO₄·7H₂O (1.16); NaH₂PO₄·H₂O (1.16); CaCl₂ (2.5); NaCl (115.5); NaHCO₃ (21.9), and glucose (11.1) at room temperature, and transported to the laboratory, where they were transferred to fresh PSS and gassed with 95% O₂ and 5% CO₂. Uteri were dissected longitudinally as previously described¹⁷ to yield 4 strips of approximately equal size using the entire horn. Each strip was tied at each end with size 2–0 silk suture, suspended from force transducers, and submerged in organ baths containing 15 mL PSS bubbled with 95% O₂ and 5% CO₂. A 10-mN resting tension was applied to each strip. Baths were subsequently heated to 37°C at which time the baseline tension was readjusted to 10 mN. The tissues were then challenged with 33 mmol/L KCl to standardize contractility responses as previously described.¹⁸ Tension was recorded isometrically using Grass FT-03 force displacement transducers (Astro-Med, Inc, West Warwick, Rhode Island) attached to a Grass Model 7D Polygraph (Astro-Med, Inc). Data were acquired and recorded using Windaq Data acquisition software (DATAQ Instruments, Inc, Akron, Ohio).

Pregnant Uterine Preparations

Uterine tissue was taken from timed-pregnant female Wistar rats at gestational day 20 (GD20; term 21–23 days). Dams were sacrificed by CO₂ asphyxiation and uteri were immediately extracted, placed in PSS at room temperature, and transported to the laboratory, where they were transferred to fresh PSS and gassed with 95% O₂ and 5% CO₂. Pups were removed, counted, weighed, and examined for gross morphological abnormalities. Eight parallel longitudinal strips approximately 15 mm in length and 3 mm in width were cut from the central portion of the uterine horn of each pregnant uterus. These strips were prepared for contractility experiments as described for DES-treated NP uterine preparations.

Red Raspberry Leaf Preparations

Three RRL preparations, tea, capsule, and extract, were purchased from local sources (tea: Traditional Medicinals, Sebastopol, California; capsule: Nature's Way, Springville, Utah; 35%–40% ethanol extract Gaia Herbs, Brevard, North Carolina; and St. Francis Herb Farm, Combermere, Ontario). Red raspberry leaf tea and an aqueous extract of the RRL capsule were prepared by removing the contents of either the tea bag or the capsule and adding each into boiling deionized water (dH₂O) to a concentration of 0.2 g/mL. The infusion was cooled (approximately 15 minutes) and centrifuged for 5 minutes at the fixed speed of an Eppendorf model 5413 microcentrifuge (Brinkmann Instruments Inc, Westbury, New York) to remove any particulate matter. After centrifugation, the supernatant was removed and used immediately in the assay. The commercial extract was used as purchased.

Uterine Contractility

Uterine contractility was assessed under 3 different experimental conditions. To determine the effect of RRL alone on uterine contractility, we examined the effect of cumulative additions of tea (1.0 µg/mL–4.6 mg/mL), capsule (1.0 µg/mL–4.6 mg/mL) and extract (2.2 ng/mL–10.1 µg/mL) on naive DES-treated NP uterine strips. Based on the results of the experiments with the DES-treated NP uterine preparations, only tea (1.0 µg/mL–4.6 mg/mL) was used in the pregnant uterine preparations. Additions were made in half-log

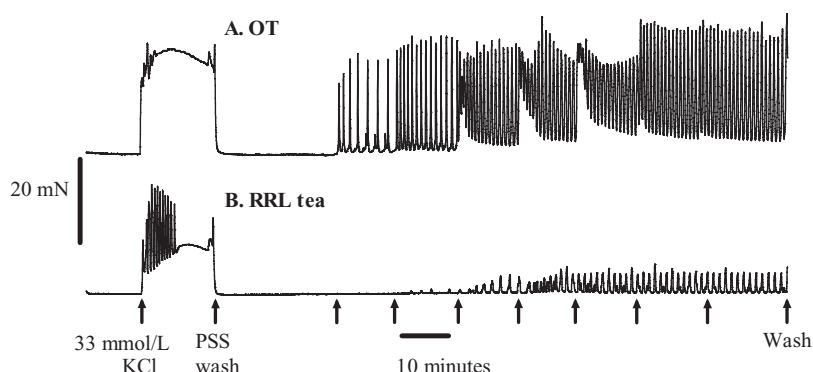


Figure 1. Representative traces of the effect of cumulative addition of (A) oxytocin and (B) red raspberry leaf (RRL) tea on the contractility of isolated strips of uterus from a diethylstilbestrol (DES)-treated nonpregnant (NP) rat. The area under the curve (AUC) recorded in the presence of 33 mmol/L KCl was used to standardize the response. The unmarked arrows indicate points where cumulative additions were made. For oxytocin, the resulting bath concentrations were 0.01, 0.04, 0.14, 0.47, 1.47, 4.7, 14.7, and 47.2 nmol/L, respectively. For RRL tea, the resulting bath concentrations were 0.001, 0.0042, 0.014, 0.046, 0.146, 0.47, 1.47, and 4.6 mg/mL, respectively. Each trace is from a different uterine strip obtained from the same animal.

concentration increments every 12 minutes, allowing 2 minutes for equilibration followed by a 10-minute recording interval.

To determine whether RRL could modify the ability of oxytocin to stimulate uterine contractility, we measured the effect of cumulative additions of oxytocin (Bachem Bioscience Inc, King of Prussia, Pennsylvania; 10 pmol/L to 0.15 μ mol/L) on uterine strips pretreated with RRL. In DES-treated NP rats, pretreatment was for 15 minutes with tea (3.2 mg/mL), capsule (3.2 mg/mL), or extract (6.9 μ g/mL). In pregnant rats, pretreatment was for 50 minutes with RRL tea (3.2 mg/mL). For pregnant animals, we used a longer preincubation time because the addition of RRL tea to the organ baths immediately stimulated uterine contractility, an effect that resolved to baseline 50 minutes after dosing.

Finally, we investigated the effect of cumulative doses of RRL preparations on uterine contractions in response to 1 nmol/L oxytocin. Oxytocin was allowed to equilibrate in the bath for 20 minutes before the cumulative additions of RRL tea (1.0 μ g/mL–4.6 mg/mL), capsule (1.0 μ g/mL–4.6 mg/mL), and extract (2.2 ng/mL–10.1 μ g/mL). Time-control experiments consisted of strips to which an equal volume of solvent was applied (tea, capsule, and oxytocin prepared in deionized water and extract in 35% ethanol). This procedure was required to compensate for any effects of vehicle or time on uterine contractility.¹⁷ At the conclusion of the experiments, 10 μ mol/L methoxyverapamil (Sigma, St Louis, Missouri) was added to the baths to relax the tissue strips to their basal tone. The value of basal tone was equated to 100% relaxation.

Data Analysis

Excitatory concentration–effect curves for RRL alone and oxytocin with RRL pretreatment. All contractility data were collected as the area under the contraction curve (AUC) for a defined 10-minute interval. For each concentration of substance (X), the percentage response obtained for the tissue treated with the test substance (E_X) was calculated using Equation 1:¹⁷

$$E_X = ((R_X - C2)/(R_{KCl} - C1)) * 100 \quad (1)$$

where R_X is the response of the tissue to substance at concentration X; $C1$, the baseline reading before the KCl challenge; R_{KCl} , the response of the uterus to KCl challenge; and $C2$, the baseline reading after the KCl challenge.

Concentration–effect curves (Figure 1) were then obtained from the data collected via the following equation:

$$E_X = \text{Max}/(1 + (10^{-(pEC_{50} - \log[X])})) \quad (2)$$

where max is the maximal response to cumulative additions of X; pEC_{50} , the negative log of the concentration of the substance required to produce 50% of the maximal response; and $\log [X]$, the log cumulative concentration of substance X.

Inhibitory Concentration–Effect Curves for RRL With Oxytocin Pretreatment

For each concentration of RRL preparation (X), the percentage of response obtained for the corresponding

time-matched control tissue (E_{CX}) was calculated using Equation 3:¹⁷

$$E_{CX} = ((C_X - C_L)/(C_F - C_L)) * 100 \quad (3)$$

where C_X is the response of the time-matched control at RRL concentration X ; C_L , the basal tone of the time-matched control tissue; and C_F , the first (control) collection from the time-matched control tissue after the addition of oxytocin.

For each concentration of RRL (X), the percentage response obtained for the tissue treated with RRL (E_{RX}) was calculated using Equation 4:¹⁷

$$E_{RX} = ((R_X - R_L)/(R_F - R_L)) * 100 \quad (4)$$

where R_X is the response of the tissue treated with RRL at concentration X ; R_L , the basal tone of the treated tissue; and R_F , the first (control) collection from the treated tissue after the addition of oxytocin.

The percentage effect produced by RRL at each concentration (E_X) was calculated using Equation 5:¹⁷

$$E_X = ((E_{RX} - E_{CX})/ - E_{CX}) * 100 \quad (5)$$

Finally, concentration–effect curves were obtained from the data collected via the following equation:

$$E_X = 100 + (-100/(1 + (10^{-(pIC_{50} - \log[X])}))) \quad (6)$$

where pIC_{50} is the negative log of the concentration of substance yielding 50% inhibition of the maximal response.

Concentration–effect curves were fitted to Equations 2 and 6 by nonlinear least squares regression using the Solver function in Microsoft Excel. Values from each animal are the average of 1 to 3 replicates. For ease of interpretation of the data, mean pEC_{50} values were converted to EC_{50} (concentration of the substance required to produce 50% of the maximal response) values according to:

$$EC_{50} = 10^{-pEC_{50}} \quad (7)$$

For the pregnant animals, only tissues from dams that did not have pups with visible morphological abnormalities were included in the analysis. Statistical significance was tested using 1-way analysis of variance (ANOVA) applying Dunn's post hoc comparison. Values of $P < .05$ were considered statistically significant.

RESULTS

Excitation Response to RRL Preparations

Representative tracings of the effects of oxytocin and RRL tea on the contractile activity of DES-treated NP uterus in vitro are shown in Figure 1. In DES-treated NP animals, both RRL tea and capsule alone caused excitation of uterine smooth muscle; however, the magnitude of the contraction was small compared to oxytocin-induced contraction. Extract had no effect. In pregnant animals, RRL tea excited uterine contractions to a degree similar to that induced by oxytocin (Figure 2, Table 1).

Effects of Pretreatment With RRL Preparations on Oxytocin Response

In DES-treated NP animals, the potency of oxytocin was unaffected by pretreatment with any formulation of RRL. In the pregnant animals, pretreatment with tea had no effect on the response to oxytocin (Table 2).

Effects of RRL Preparations on Contractions in Oxytocin-Exposed Uterine Strips

In uteri from DES-treated NP animals, none of the RRL preparations affected oxytocin-driven contractions until the highest concentrations used, at which uterine contraction was partially reduced (Table 3). The ability of tea to alter uterine contractility in pregnant animals was variable. Red raspberry leaf augmented oxytocin-driven excitation in some animals ($n = 3$) and produced a biphasic response (initial excitation followed by inhibition) in others ($n = 3$; Figure 3). In all cases, the same type of response was observed in different strips from the same animal.

DISCUSSION

Results from this study have demonstrated that RRL tea stimulated contractions in uterine strips taken from DES-treated NP animals and GD20 pregnant animals. In most instances (see Figure 1 for example), the strips were quiescent until the addition of the stimulant; this is the usual case when the methods we employed are used.^{17,19} Previous work by ourselves²⁰ and others²¹ has revealed that maximal responses of isolated rat uteri are quite variable from animal to animal, particularly in late pregnancy.

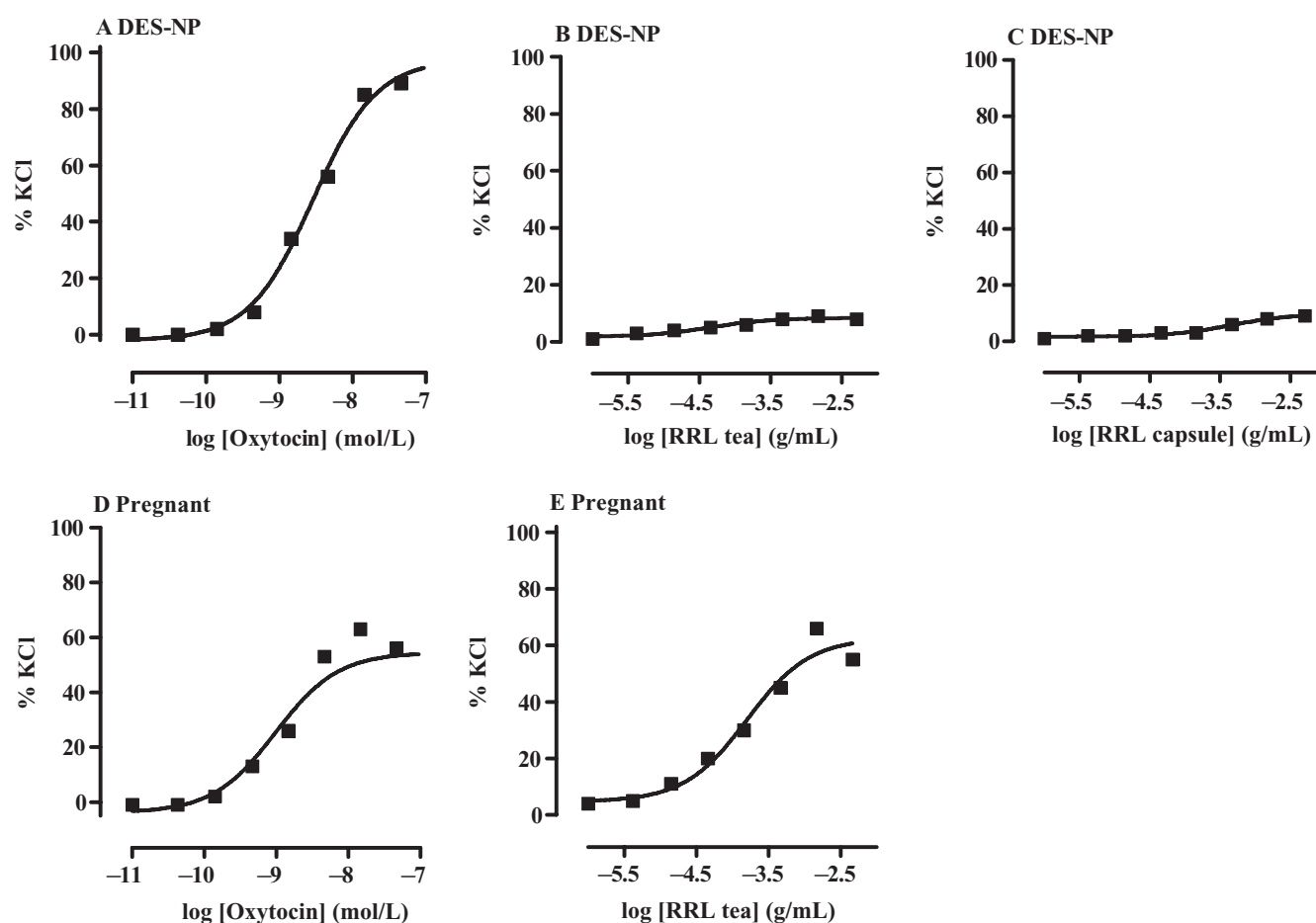


Figure 2. Representative concentration–effect curves for oxytocin (A and D), red raspberry leaf (RRL) tea (B and E), and RRL capsule (C) alone on the contractility of the uterus from diethylstilbestrol (DES)–treated nonpregnant (NP; A–C) and day 20 pregnant rats (D and E). Points are means of duplicates in a single experiment.

Table 1. Effects of Test Substances Alone on Rat Uterine Contractility In Vitro^a

Status	Treatment	EC ₅₀	pEC ₅₀	Max (% KCl)	n
DES-NP	Oxytocin	5.0 nM	8.3 ± 0.1 (–log M)	113 ± 18	4
	RRL tea	0.4 mg/mL	3.4 ± 0.4 (–log g/mL)	18 ± 3 ^b	4
	RRL capsule	0.6 mg/mL	3.2 ± 0.4 (–log g/mL)	8.6 ± 2 ^b	3
	RRL extract		No effect	0 ^b	3
Pregnant	Oxytocin	1.6 nM	8.8 ± 0.2 (–log M)	74 ± 19	5
	RRL Tea	0.2 mg/mL	3.7 ± 0 (–log g/mL)	45 ± 17	6

Abbreviations: DES, diethylstilbestrol; EC₅₀, concentration of the substance required to produce 50% of the maximal response; NP, nonpregnant; pEC₅₀, the negative log of the concentration of the substance required to produce 50% of the maximal response; RRL, red raspberry leaf.

^a Values are means ± SEM from n experiments.

^b Maximum responses to RRL were significantly different from those to oxytocin in DES-treated NP but not in pregnant rats.

Such was the case in the current study (Tables 1 and 2), nevertheless, we were able to show that in the DES-treated NP uterus, the degree of excitation was significantly lower than that seen with oxytocin, whereas RRL

tea stimulated contractions to a similar degree as oxytocin in uterine strips taken from GD20 pregnant animals. Moreover, in 50% of pregnant animals, the cotreatment of tea and oxytocin enhanced the contractility of uterine

Table 2. Effects of Pretreatment With RRL Preparations on the Concentration–Effect Relationship to Oxytocin in Rat Uterine Strips^a

Status	Pretreatment	EC ₅₀ oxytocin (nM)	pEC ₅₀ oxytocin (mol/L)	Max (% KCl)	n
DES-NP (water extracts)	Control	4.0	8.4 ± 0.1	166 ± 31	3
	Tea (3.2 mg/mL)	3.2	8.5 ± 0.1	136 ± 29	3
	Capsule (3.2 mg/mL)	4.0	8.4 ± 0.1	130 ± 28	3
DES-NP (ethanol extract)	Control	1.6	8.8 ± 0.1	155 ± 63	3
	Extract (6.9 µg/mL)	4.0	8.4 ± 0.1	68 ± 12	4
Pregnant	Control	1.6	8.8 ± 0.1	124 ± 23	3
	Tea (3.2 mg/mL)	1.3	8.9 ± 0.1	99 ± 10	3

Abbreviations: DES, diethylstilbestrol; EC₅₀, concentration of the substance required to produce 50% of the maximal response; NP, nonpregnant; pEC₅₀, the negative log of the concentration of the substance required to produce 50% of the maximal response; RRL, red raspberry leaf.

^a Mean ± SEM from *n* experiments.

Table 3. Effect of High Concentrations of RRL Preparations on Oxytocin-Induced (1 nM) Contractions of Uteri From DES-Treated NP Animals

Treatment	Effective Concentration	% Control (Oxytocin)	n
RRL tea	4.6 mg/mL	71 ± 16	6
RRL capsule	4.6 mg/mL	78 ± 16	6
RRL tincture	10.1 µg/mL	24 ± 7	7

Abbreviation: RRL, red raspberry leaf.

strips above that seen with oxytocin alone. The findings of this study are in contrast to those reported by Bamford et al,¹⁴ who observed that an aqueous extract of raspberry leaf did not affect the *in vitro* contractility of the nonpregnant uterus and inhibited contractions in uteri taken from pregnant animals. However, the work of Bamford et al¹⁴ has only appeared in abstract form, the nonpregnant animals were not hormonally standardized by DES treatment, and it is not clear whether the pregnant tissues were taken from all animals at the same gestational age. Uterine contractility is highly dependent on estrogen and progesterone levels and gestational age²²; therefore, the lack of standardization of the timing of tissue collection could explain the differences between our results and those of Bamford et al.¹⁴ It is interesting to note that we did observe RRL-induced inhibition of uterine contractility in tissues from 1 pregnant animal, but as this animal had 3 of 14 pups with gross morphological abnormalities, it was not included in the analysis (data not shown). In addition to differences in the characteristics of the animals used in this study versus those reported by Bamford et al,¹⁴ it is also possible that the type of RRL preparation may explain differences in the uterine responsiveness to RRL treatment.

In this study, we have demonstrated that the aqueous extracts of RRL can induce contractions in uterine strips from both pregnant (tea) and DES-treated NP animals (tea and capsule). However, the ethanol extract had no effect. This is not entirely surprising because it has previously been demonstrated that the ability of raspberry leaf extracts to affect smooth muscle contractility differs depending on the solvent used for the extraction.^{16,23} Thus, even though raspberry leaf contains components that potentially relax gastrointestinal smooth muscle *in vitro*,^{23,24} such an effect was not seen on uterine smooth muscle in the current study when commercial RRL tea was prepared in a manner similar to that recommended for therapeutic purposes. It is still unknown how RRL can affect uterine contractility, but this may depend on the active constituents in each preparation.

Raspberry leaf contains numerous phytochemicals including tannins, phenolic acids, terpenoids, and flavonoids and the extraction of these components depends on their polarity and the solvent used.^{24,25} It is unclear from the current study which constituents of the RRL preparations are responsible for causing the excitation in the pregnant and DES-treated NP uterus, but the literature suggests that it is unlikely to be the flavonoids. Indeed, a number of flavonoids including kaempferol and quercetin, both of which are present in RRL extracts,^{24,25} have been shown to cause relaxation of rat uterine smooth muscle.^{26–28} Therefore, the component of RRL that enhances uterine contractility remains to be identified.

In summary, we have demonstrated in this study that commercial preparations of RRL can affect uterine contractility in the pregnant and DES-treated NP uterus. However, the effects of RRL were only seen at the highest concentrations examined: concentrations that are

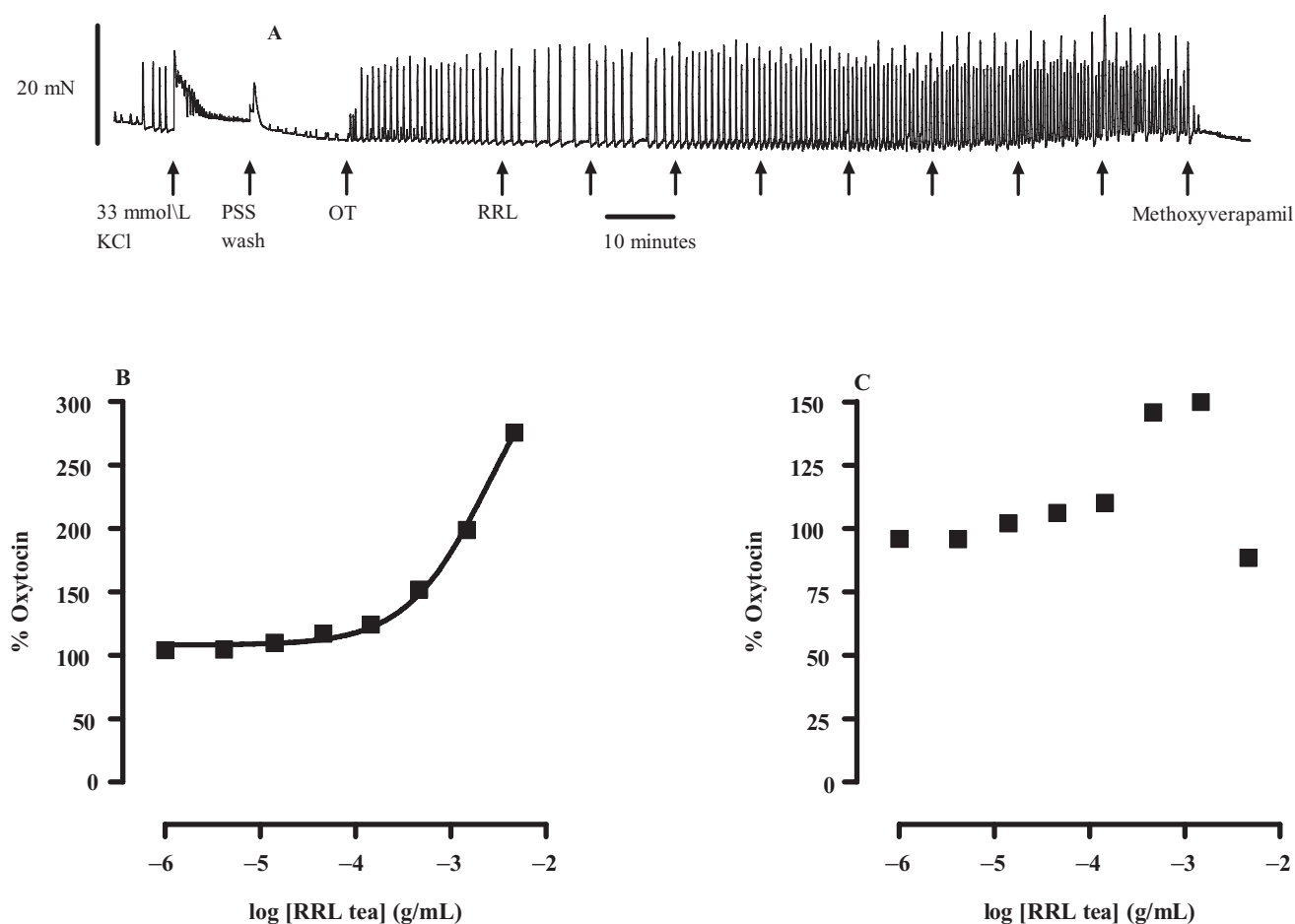


Figure 3. The effect of red raspberry leaf (RRL) tea on the contractility of uterine strips exposed to 1 nmol/L oxytocin. Strips were excised from animals at day 20 of pregnancy. Representative trace of a complete experiment (A). The unmarked arrows indicate points where cumulative additions were made. The resulting bath concentrations of RRL tea were 0.001, 0.0042, 0.014, 0.046, 0.146, 0.47, 1.47, and 4.6 mg/mL, respectively. Two types of concentration–response relationships were observed: excitation ($n = 3$; B) and biphasic excitation followed by inhibition ($n = 3$; C). Data in B and C are from individual strips from different experiments.

unlikely to be obtained through consumption of these products. In the best case, where the active component is fully bioavailable and its distribution is restricted to the plasma volume (4 L), a single cup of RRL tea would yield a plasma concentration of 0.44 mg/mL. Although this concentration is close to the EC_{50} for contraction (Table 1), the best case is unlikely to pertain, given the generally poor oral bioavailability of phytochemicals as discussed by Unger and Frank.²⁹ Therefore, the ability of RRL to reduce the incidence of forceps delivery as described by Simpson et al¹³ does not appear to be via a direct action on uterine contractility. Our data do not rule out the possibility of either a direct effect of an active metabolite of RRL on uterine contractility or an effect of RRL on other targets in the cascade of events leading to parturition such as the hypothalamic–pituitary–adrenal

axis or cervical dilatation. Indeed, animals that were given RRL throughout pregnancy had a significant increase in gestational length suggesting an effect on the timing of labor.³⁰ However, the lack of convincing biological evidence for the efficacy of this herbal preparation for stimulating labor and dearth of information regarding the safety of its use during pregnancy³¹ raises serious concerns about whether its use should be recommended for pregnant women.

ACKNOWLEDGMENTS

We thank Ms Priyanka Silveira Da Graca Costa, Mr Anton Shimanovsky, and the staff of the Central Animal Facility at McMaster University for their excellent technical assistance. This project was supported by

a SickKids Foundation grant in Complementary and Alternative Health Care and Paediatrics and the Canadian Institutes of Health Research (ACH and DJC).

REFERENCES

- Kelly JP, Kaufman DW, Kelley K, Rosenberg L, Mitchell AA. Use of herbal/natural supplements according to racial/ethnic group. *J Altern Complement Med*. 2006;12(6):555-561.
- Najm W, Lie D. Dietary supplements commonly used for prevention. *Prim Care*. 2008;35(4):749-767.
- Eisenberg DM, Davis RB, Ettner SL, et al. Trends in alternative medicine use in the United States, 1990-1997: results of a follow-up national survey. *JAMA*. 1998;280(18):1569-1575.
- Kennedy J. Herb and supplement use in the US adult population. *Clin Ther*. 2005;27(11):1847-1858.
- Conover EA. Herbal agents and over-the-counter medications in pregnancy. *Best Pract Res Clin Endocrinol Metab*. 2003;17(2):237-251.
- Holst L, Wright D, Haavik S, Nordeng H. The use and the user of herbal remedies during pregnancy. *J Altern Complement Med*. 2009;15(7):787-792.
- Nordeng H, Havnen GC. Impact of socio-demographic factors, knowledge and attitude on the use of herbal drugs in pregnancy. *Acta Obstet Gynecol Scand*. 2005;84(1):26-33.
- Refuerzo JS, Blackwell SC, Sokol RJ, et al. Use of over-the-counter medications and herbal remedies in pregnancy. *Am J Perinatol*. 2005;22(6):321-324.
- Tsui B, Dennehy CE, Tsourounis C. A survey of dietary supplement use during pregnancy at an academic medical center. *Am J Obstet Gynecol*. 2001;185(2):433-437.
- Tiran D. The use of herbs by pregnant and childbearing women: a risk-benefit assessment. *Complement Ther Nurs Midwifery*. 2003;9(4):176-181.
- Low Dog T. The use of botanicals during pregnancy and lactation. *Altern Ther Health Med*. 2009;15(1):54-58.
- McFarlin BL, Gibson MH, O'Rear J, Harman P. A national survey of herbal preparation use by nurse-midwives for labor stimulation. Review of the literature and recommendations for practice. *J Nurse Midwifery*. 1999;44(3):205-216.
- Simpson M, Parsons M, Greenwood J, Wade K. Raspberry leaf in pregnancy: its safety and efficacy in labor. *J Midwifery Womens Health*. 2001;46(2):51-59.
- Bamford DS, Percival RC, Tothill AU. Raspberry leaf tea: a new aspect to an old problem. *Br J Pharmacol*. 1970;40(1):161P-162P.
- Burn JH, Withell ER. A principle in raspberry leaves which relaxes uterine muscle. *Lancet*. 1941;238(July 5):1-3.
- Beckett AH, Belthle FW, Fell KR, Lockett MF. The active constituents of raspberry leaves; a preliminary investigation. *J Pharm Pharmacol*. 1954;6(11):785-796.
- Crankshaw DJ. Pharmacological techniques for the in vitro study of the uterus. *J Pharmacol Toxicol Methods*. 2001;45(2):123-140.
- Niuro N, Nishimura J, Hirano K, Nakano H, Kanaide H. Mechanisms of galanin-induced contraction in the rat myometrium. *Br J Pharmacol*. 1998;124(8):1623-1632.
- Wainman BC, Burcea I, Crankshaw DJ. The effects of prostanoids on estrogen dominated rat myometrial longitudinal muscle. *Biol Reprod*. 1988;39(2):221-228.
- Crankshaw DJ, Gaspar V. Effects of prostanoids on the rat's myometrium in vitro during pregnancy. *Biol Reprod*. 1992;46(3):392-400.
- Freerksen N, Betancourt A, Maul H, et al. PAR-2 activating peptide-induced stimulation of pregnant rat myometrium contractile activity partly involves the other membrane receptors. *Eur J Obstet Gynecol Reprod Biol*. 2007;130(1):51-59.
- Mesiano S, Welsh TN. Steroid hormone control of myometrial contractility and parturition. *Semin Cell Dev Biol*. 2007;18(3):321-331.
- Rojas-Vera J, Patel AV, Dacke CG. Relaxant activity of raspberry (*Rubus idaeus*) leaf extract in guinea-pig ileum in vitro. *Phytother Res*. 2002;16(7):665-668.
- Patel AV, Rojas-Vera J, Dacke CG. Therapeutic constituents and actions of *Rubus* species. *Curr Med Chem*. 2004;11(11):1501-1512.
- Gudej J, Tomczyk M. Determination of flavonoids, tannins and ellagic acid in leaves from *Rubus* L. species. *Arch Pharm Res*. 2004;27(11):1114-1119.
- Revuelta MP, Cantabrana B, Hidalgo A. Depolarization-dependent effect of flavonoids in rat uterine smooth muscle contraction elicited by CaCl₂. *Gen Pharmacol*. 1997;29(5):847-857.
- Revuelta MP, Hidalgo A, Cantabrana B. Involvement of cAMP and beta-adrenoceptors in the relaxing effect elicited by flavonoids on rat uterine smooth muscle. *J Auton Pharmacol*. 1999;19(6):353-358.
- Revuelta MP, Cantabrana B, Hidalgo A. Mechanisms involved in kaempferol-induced relaxation in rat uterine smooth muscle. *Life Sci*. 2000;67(3):251-259.
- Unger M, Frank A. Simultaneous determination of the inhibitory potency of herbal extracts on the activity of six major cytochrome P450 enzymes using liquid chromatography / mass spectrometry and automated online extraction. *Rapid Commun Mass Spectrom*. 2004;18(19):2273-81.
- Johnson JR, Makaji E, Ho S, Xiong B, Crankshaw DJ, Holloway AC. Effect of maternal raspberry leaf consumption in rats on pregnancy outcome and the fertility of the female offspring. *Reprod Sci*. 2009;16(6):605-609.
- Holst L, Haavik S, Nordeng H. Raspberry leaf—should it be recommended to pregnant women? *Complement Ther Clin Pract*. 2009;15(4):204-208.